

3.1

$$(a) \begin{matrix} 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 0 \\ 2^9 & 2^8 & 2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \end{matrix} = (478)_{10}$$

$$(b) \begin{matrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 & 1 & 1 & 1 \\ 2^9 & 2^8 & 2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \end{matrix} = (443)_{10}$$

$$(c) (3751)_8 = (2025)_{10}$$

$$(d) (A225F)_{16} = (664159)_{10}$$

$$(e) (FOFO)_{16} = (61680)_{10}$$

3.2

$$(a) (0111011110)_2 = (478)_{10}$$

$$(1011100111)_2 = (280)_{10}$$

$$(1111111110) = (-1)_{10}$$

3.3

$$(a) 0111011110 = 478$$

$$(b) 1011100111 = -281$$

$$(c) 1111111110 = -2$$

3.4 ~~73, 4506, -85~~

34

73

(a) 01001001

(b) 01001001

(c) 01001001

-85

(a) ~~1~~101111

(b) 11010000

(c) 110100001

1806

(a) 011101110010

(b) 011101110010

(c) 011101110010

-1830

(a) 111001011110

(b) 100110100001

(c) 100110100010